

Simple Linear Regression

Statistics App

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Table of contents

1	Sum of Squares	2
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1 Sum of Squares

$$SXX = \sum_{i=1}^n x_i(x_i - \bar{x}) \quad (1)$$

```
class OLS:
    def __init__(self):
        pass

    def mean(self, x):
        n = len(x)
        value_mean = sum(i for i in range(n))
        return (1 / n) * value_mean

    def sumOfSquares(self, x):
        value_mean = self.mean(x)
        value_ss = sum(x[i] * (x[i] - value_mean) for i in range(len(x)))
        return value_ss

var = OLS()
x_val = [1, 2, 3]
var.sumOfSquares(x_val)
```

8.0